

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 th June 2026
Student Name	Tirumala Srinidhi
Project Name	ShopEZ – E-Commerce Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form, Gmail, Facebook
FR-2	User Confirmation	Confirmation via Email, Confirmation via OTP
FR-3	Product Browsing & Search	View product catalog, Apply filters, Read reviews
FR-4	Cart Management	Add products to cart, Remove products, View cart summary
FR-5	Order Placement & Checkout	Enter address, Select payment method, Place order
FR-6	Order Confirmation & Tracking	Receive confirmation notification, View order details
FR-7	Admin Dashboard	Manage products, Categories, Banners, View analytics

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be intuitive and responsive across devices, ensuring smooth navigation for users.
NFR-2	Security	All user data, including payment details, must be encrypted and protected using secure protocols (HTTPS, JWT).
NFR-3	Reliability	The system should ensure consistent performance with minimal downtime and accurate order processing.
NFR-4	Performance	Pages should load within 3 seconds, and backend APIs should respond efficiently under high traffic.
NFR-5	Availability	The platform should maintain 99.9% uptime to ensure continuous access for users and sellers.
NFR-6	Scalability	The system should support increasing numbers of users, products, and orders without performance degradation.